# AiDyHD — ADHD Diagnostic Analytics Using PostgreSQL

**Clinical Data Analysis Report (SQL)**

# Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder that affects attention, impulse control, and behavioral regulation. Understanding the behavioral patterns and lifestyle factors associated with ADHD can help in early detection and intervention.

This project analyzes a dataset containing 6,500 patient records ranging from ages 3 to 55. The analysis is conducted using PostgreSQL, leveraging advanced SQL features such as window functions, common table expressions (CTEs), statistical functions, and cohort analysis techniques.

The objective of this study is to identify patterns in ADHD diagnosis and evaluate how demographics, lifestyle habits, and behavioral indicators influence ADHD severity and prevalence.

# Dataset Description

The dataset consists of 32 variables capturing demographic, behavioral, and clinical characteristics.

### Demographic Variables

- Age
- Gender
- Educational Level
- Family History of ADHD

### Lifestyle Variables

- Sleep Hours
- Daily Phone Usage
- Daily Physical Activity
- Coffee and Tea Consumption

## Behavioral Questionnaire

**Two sets of behavioral questionnaires were used:**

**Hyperactivity Questions**

- Q1_1 to Q1_9

**Inattention Questions**

- Q2_1 to Q2_9

**These responses are aggregated to create behavioral indices.**

## Derived Metrics

**Several calculated variables were created:**

- Hyperactivity Index
- Inattention Index
- Total Severity Score
- Sleep Risk Flag
- High Screen Time Flag
- Anxiety Risk Flag

## ADHD Diagnosis Classes

| Code | Diagnosis Type |
|---|---|
| 0 | No ADHD |
| 1 | Inattentive Type |
| 2 | Hyperactive Type |
| 3 | Combined Type |

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# 1. Database Setup and Data Cleaning

The first step involved creating the main database table and importing the raw CSV dataset into PostgreSQL.

The view performs the following transformations:

- Converts gender codes into readable labels.
- Maps diagnosis codes into descriptive labels.
- Calculates hyperactivity and inattention scores.
- Computes total ADHD severity index.
- Categorizes individuals into life stages.
- Creates lifestyle risk flags for further analysis.

These transformations allow the dataset to be used efficiently for analytical queries.

Output Screenshot:

ADHD local: SELECT * FROM a...

| id | age | gender | educational_level | family_history | sleep_hours |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| 1 | 8 | 1 | Primary | No | 8 |
| 2 | 9 | 2 | Primary | No | 11 |
| 3 | 9 | 1 | Primary | No | 9 |
| 4 | 5 | 2 | Kindergarten | Yes | 7 |
| 5 | 13 | 1 | Middle | No | 3 |
| 6 | 5 | 1 | Kindergarten | No | 9 |
| 7 | 7 | 1 | Primary | No | 7 |
| 8 | 7 | 1 | Primary | No | 7 |
| 9 | 12 | 2 | Primary | No | 6 |
| 10 | 3 | 1 | Kindergarten | No | 8 |

# 2. Cohort Overview Analysis

This analysis provides a summary of the dataset population.

Key metrics calculated include:

- Total number of patients
- Average patient age
- Minimum and maximum age
- ADHD prevalence rate
- Gender distribution

Additionally, patients are categorized into life stages:

- Early Childhood
- Pediatric
- Adolescent
- Adult

This helps understand how ADHD prevalence varies across different age groups.

Output Screenshot:

2a. Full dataset overview

| total_patients | avg_age | min_age | max_age | total_adhd | overall_prevalen... |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| 6500 | 11.5 | 3 | 55 | 4900 | 75.4 |

| male_count | female_count |
|---|---|
| Filter... | Filter... |
| 3900 | 2600 |

2b. Overview by life stage

| life_stage | patients | pct_of_total | adhd_count | adhd_prevalenc... | avg_total_severity |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| Early Childhood (3-4) | 846 | 13.0 | 635 | 75.1 | 30.17 |
| Pediatric (5-14) | 4121 | 63.4 | 3115 | 75.6 | 30.41 |
| Adolescent (15-18) | 883 | 13.6 | 669 | 75.8 | 30.39 |
| Adult (19-55) | 650 | 10.0 | 481 | 74.0 | 29.79 |

# 3. Diagnosis Distribution

This section analyzes the distribution of ADHD diagnosis types across the dataset.

The analysis measures:

- Number of patients per diagnosis type
- Percentage share of each diagnosis
- Average symptom severity
- Hyperactivity and inattention score averages

This helps determine which ADHD subtype is most common within the population.

Output Screenshot:

3a. Full dataset diagnosis distribution

| diagnosis_label | patient_count | percentage | avg_severity | avg_hyperactivity | avg_inattention |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| No ADHD | 1600 | 24.6 | 8.98 | 4.52 | 4.46 |
| Inattentive Type | 1050 | 16.2 | 26.98 | 4.47 | 22.52 |
| Hyperactive Type | 1050 | 16.2 | 26.96 | 22.50 | 4.46 |
| Combined Type | 2800 | 43.1 | 45.00 | 22.51 | 22.50 |

3b. Pediatric cohort only (5-14) — your Excel dashboard cohort

| diagnosis_label | patient_count | percentage |
|---|---|---|
| Filter... | Filter... | Filter... |
| No ADHD | 1006 | 24.4 |
| Inattentive Type | 659 | 16.0 |
| Hyperactive Type | 664 | 16.1 |
| Combined Type | 1792 | 43.5 |

# 4. ADHD Prevalence by Age

This query analyzes how ADHD prevalence changes with age.

Metrics calculated include:

- Number of patients at each age
- ADHD diagnosis count
- ADHD prevalence percentage
- 3-year rolling prevalence average
- cumulative patient counts

A window function is used to compute the rolling average, which smooths fluctuations and helps observe long-term trends.

Output Screenshot:
*Window function — shows symptom trajectory across childhood*

| age | total_patients | adhd_count | prevalence_pct | rolling_3yr_prev... | cumulative_pati... |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| 5 | 445 | 332 | 74.6 | 74.6 | 445 |
| 6 | 444 | 350 | 78.8 | 76.7 | 889 |
| 7 | 449 | 335 | 74.6 | 76.0 | 1338 |
| 8 | 467 | 360 | 77.1 | 76.8 | 1805 |
| 9 | 434 | 336 | 77.4 | 76.4 | 2239 |
| 10 | 507 | 370 | 73.0 | 75.8 | 2746 |
| 11 | 450 | 341 | 75.8 | 75.4 | 3196 |
| 12 | 508 | 377 | 74.2 | 74.3 | 3704 |
| 13 | 203 | 154 | 75.9 | 75.3 | 3907 |
| 14 | 214 | 160 | 74.8 | 74.9 | 4121 |
| 15 | 221 | 168 | 76.0 | 75.5 | 4342 |
| 16 | 214 | 166 | 77.6 | 76.1 | 4556 |
| 17 | 217 | 161 | 74.2 | 75.9 | 4773 |
| 18 | 231 | 174 | 75.3 | 75.7 | 5004 |

# 5. Life Stage Cohort Comparison

This analysis compares lifestyle patterns and behavioral indicators across different life stages.

Variables analyzed include:

- Sleep duration
- Screen time
- Physical activity
- Caffeine consumption
- Anxiety levels
- Learning difficulties

The results help identify how ADHD symptoms manifest differently across age groups.

Output Screenshot:

*ADHD vs No-ADHD across all lifestyle variables — per life stage*

| life_stage | diagnosis_label | n | avg_severity | avg_sleep | avg_screen_time |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| Adolescent (15-18) | No ADHD | 214 | 8.97 | 7.71 | 3.66 |
| Adolescent (15-18) | Inattentive Type | 140 | 27.33 | 7.68 | 3.68 |
| Adolescent (15-18) | Hyperactive Type | 152 | 27.03 | 7.51 | 3.42 |
| Adolescent (15-18) | Combined Type | 377 | 45.03 | 7.59 | 3.41 |
| Adult (19-55) | No ADHD | 169 | 8.98 | 7.52 | 3.43 |
| Adult (19-55) | Inattentive Type | 108 | 26.94 | 7.80 | 3.61 |
| Adult (19-55) | Hyperactive Type | 101 | 26.79 | 7.58 | 3.70 |
| Adult (19-55) | Combined Type | 272 | 44.96 | 7.53 | 3.52 |
| Early Childhood (3-4) | No ADHD | 211 | 9.02 | 7.55 | 3.57 |
| Early Childhood (3-4) | Inattentive Type | 143 | 26.82 | 7.24 | 3.76 |
| Early Childhood (3-4) | Hyperactive Type | 133 | 26.95 | 7.39 | 3.53 |
| Early Childhood (3-4) | Combined Type | 359 | 45.13 | 7.62 | 3.46 |
| Pediatric (5-14) | No ADHD | 1006 | 8.97 | 7.46 | 3.62 |
| Pediatric (5-14) | Inattentive Type | 659 | 26.95 | 7.51 | 3.48 |
| Pediatric (5-14) | Hyperactive Type | 664 | 26.98 | 7.51 | 3.61 |
| Pediatric (5-14) | Combined Type | 1792 | 44.98 | 7.50 | 3.56 |

| avg_activity | avg_caffeine | avg_focus_score | avg_anxiety | avg_learning_diff | avg_org_difficulty |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| 1.02 | 2.00 | 5.42 | 0.65 | 0.30 | 0.27 |
| 0.95 | 1.95 | 5.14 | 1.21 | 0.21 | 0.26 |
| 1.16 | 1.80 | 3.54 | 1.24 | 0.72 | 1.00 |
| 1.00 | 1.91 | 3.53 | 1.27 | 0.74 | 1.00 |
| 0.93 | 2.11 | 5.31 | 0.49 | 0.21 | 0.30 |
| 0.93 | 2.01 | 5.61 | 1.28 | 0.29 | 0.29 |
| 0.93 | 1.99 | 3.88 | 1.29 | 0.80 | 1.00 |
| 1.04 | 1.91 | 3.44 | 1.22 | 0.79 | 1.00 |
| 1.06 | 0.26 | 5.56 | 0.70 | 0.25 | 0.31 |
| 1.11 | 0.28 | 5.48 | 1.18 | 0.22 | 0.36 |
| 1.07 | 0.20 | 3.68 | 1.10 | 0.74 | 1.00 |
| 1.09 | 0.17 | 3.80 | 1.34 | 0.75 | 1.00 |
| 1.02 | 0.41 | 5.49 | 0.67 | 0.23 | 0.30 |
| 1.01 | 0.38 | 5.50 | 1.17 | 0.25 | 0.32 |
| 1.00 | 0.41 | 3.60 | 1.26 | 0.77 | 1.00 |
| 1.03 | 0.37 | 3.62 | 1.21 | 0.78 | 1.00 |

# 6. Risk Stratification Model

A multi-stage Common Table Expression (CTE) pipeline is used to classify patients into risk categories.

Risk categories include:

- Low Risk
- Moderate Risk
- High Risk
- Critical Risk

The classification is based on:

- Sleep deprivation
- excessive screen time
- anxiety and depression levels

This model provides insight into how lifestyle and mental health indicators contribute to ADHD severity.

Output Screenshot:
3-stage CTE pipeline: score → flag → summarise

| diagnosis_label | risk_category | patient_count | avg_severity | avg_sleep | avg_screen_time |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| Combined Type | Critical Risk | 26 | 45.42 | 5.35 | 5.92 |
| Combined Type | High Risk | 135 | 45.05 | 5.21 | 5.87 |
| Combined Type | Moderate Risk | 771 | 45.00 | 6.89 | 4.18 |
| Combined Type | Low Risk | 860 | 44.93 | 8.47 | 2.58 |
| Hyperactive Type | Critical Risk | 11 | 27.09 | 5.73 | 5.73 |
| Hyperactive Type | High Risk | 60 | 26.35 | 5.50 | 5.78 |
| Hyperactive Type | Moderate Risk | 269 | 27.08 | 6.92 | 4.16 |
| Hyperactive Type | Low Risk | 324 | 27.00 | 8.44 | 2.68 |
| Inattentive Type | Critical Risk | 13 | 26.85 | 5.38 | 5.46 |
| Inattentive Type | High Risk | 35 | 27.37 | 5.31 | 5.51 |
| Inattentive Type | Moderate Risk | 292 | 26.91 | 6.90 | 4.10 |
| Inattentive Type | Low Risk | 319 | 26.95 | 8.40 | 2.61 |
| No ADHD | Critical Risk | 5 | 8.80 | 5.20 | 6.20 |
| No ADHD | High Risk | 89 | 9.01 | 5.10 | 5.91 |
| No ADHD | Moderate Risk | 418 | 8.94 | 6.92 | 4.28 |
| No ADHD | Low Risk | 494 | 8.98 | 8.36 | 2.63 |

| avg_anxiety | family_history_c... | pct_within_diag... |
|---|---|---|
| Filter... | Filter... | Filter... |
| 3.00 | 3 | 1.5 |
| 0.90 | 17 | 7.5 |
| 1.23 | 163 | 43.0 |
| 1.19 | 167 | 48.0 |
| 3.00 | 1 | 1.7 |
| 0.88 | 9 | 9.0 |
| 1.31 | 53 | 40.5 |
| 1.23 | 52 | 48.8 |
| 3.00 | 6 | 2.0 |
| 0.86 | 4 | 5.3 |
| 1.12 | 62 | 44.3 |
| 1.16 | 65 | 48.4 |
| 3.00 | 1 | 0.5 |
| 0.56 | 21 | 8.8 |
| 0.63 | 88 | 41.6 |
| 0.71 | 82 | 49.1 |

# 7. Statistical Percentile Analysis

Statistical functions were used to calculate percentile distributions for behavioral indices.

Metrics computed include:

- Mean values
- Standard deviation
- 25th percentile
- Median
- 75th percentile
- 90th percentile

This provides a deeper understanding of the distribution of symptom severity across diagnosis groups.

Output Screenshot:
Produces clinical-grade score distribution table

| p75_hyperactivity | mean_inattention | sd_inattention | p25_inattention | median_inattent... | p75_inattention |
|---|---|---|---|---|---|
| abc Filter... | abc Filter... | abc Filter... | abc Filter... | abc Filter... | abc Filter... |
| 6 | 4.46 | 1.51 | 3 | 4 | 6 |
| 5 | 22.52 | 1.50 | 22 | 22 | 24 |
| 23 | 4.46 | 1.52 | 3 | 4 | 5 |
| 24 | 22.50 | 1.53 | 21 | 23 | 24 |


| p75_hyperactivity | mean_inattention | sd_inattention | p25_inattention | median_inattent... | p75_inattention |
|---|---|---|---|---|---|
| abc Filter... | abc Filter... | abc Filter... | abc Filter... | abc Filter... | abc Filter... |
| 6 | 4.46 | 1.51 | 3 | 4 | 6 |
| 5 | 22.52 | 1.50 | 22 | 22 | 24 |
| 23 | 4.46 | 1.52 | 3 | 4 | 5 |
| 24 | 22.50 | 1.53 | 21 | 23 | 24 |


| mean_total_seve... | median_total_se... | p90_total_severity |
|---|---|---|
| abc Filter... | abc Filter... | abc Filter... |
| 8.98 | 9 | 12 |
| 26.98 | 27 | 30 |
| 26.96 | 27 | 30 |
| 45.00 | 45 | 48 |

# 8. Lifestyle Factor Analysis

Lifestyle behaviors were analyzed to understand their correlation with ADHD severity.

Two primary factors were studied:

## Sleep Patterns

Patients were grouped based on sleep duration categories to measure its effect on ADHD severity.

## Screen Time

Patients were grouped by daily phone usage to observe changes in:

- ADHD prevalence
- anxiety levels
- behavioral severity scores

Output Screenshot:

8a. Sleep hours distribution across diagnosis groups

| diagnosis_label | sleep_category | patient_count | avg_severity |
|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... |
| No ADHD | Under 6hrs (Severe Sleep Risk) | 115 | 8.93 |
| No ADHD | 6-7hrs (Mild Sleep Risk) | 179 | 8.96 |
| No ADHD | 7-9hrs (Normal) | 426 | 8.91 |
| No ADHD | Over 9hrs | 286 | 9.07 |
| Inattentive Type | Under 6hrs (Severe Sleep Risk) | 67 | 26.82 |
| Inattentive Type | 6-7hrs (Mild Sleep Risk) | 132 | 27.08 |
| Inattentive Type | 7-9hrs (Normal) | 268 | 26.83 |
| Inattentive Type | Over 9hrs | 192 | 27.08 |
| Hyperactive Type | Under 6hrs (Severe Sleep Risk) | 65 | 27.23 |
| Hyperactive Type | 6-7hrs (Mild Sleep Risk) | 138 | 26.73 |
| Hyperactive Type | 7-9hrs (Normal) | 264 | 27.00 |
| Hyperactive Type | Over 9hrs | 197 | 27.04 |
| Combined Type | Under 6hrs (Severe Sleep Risk) | 183 | 44.95 |
| Combined Type | 6-7hrs (Mild Sleep Risk) | 371 | 44.99 |
| Combined Type | 7-9hrs (Normal) | 703 | 45.01 |
| Combined Type | Over 9hrs | 535 | 44.95 |

8b. Screen time vs severity — ranked buckets

| screen_time_buc... | total_patients | avg_severity | avg_anxiety | adhd_count | adhd_prevalenc... |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| Low (<2hrs) | 667 | 30.91 | 1.13 | 511 | 76.6 |
| Moderate (2-4hrs) | 1393 | 30.31 | 1.07 | 1054 | 75.7 |
| High (4-6hrs) | 1419 | 30.31 | 1.07 | 1077 | 75.9 |
| Very High (6+ hrs) | 642 | 30.30 | 1.09 | 473 | 73.7 |

# 9. Family History Impact

This analysis evaluates the relationship between family history and ADHD diagnosis.

Metrics include:

- ADHD prevalence among individuals with family history
- severity score differences
- hyperactivity and inattention score comparisons

The results help identify possible genetic influences on ADHD.

Output Screenshot:
Does family history of ADHD predict higher severity?

| family_history | diagnosis_label | patient_count | avg_severity | avg_hyperactivity | avg_inattention |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| No | No ADHD | 718 | 8.91 | 4.44 | 4.47 |
| No | Inattentive Type | 440 | 27.08 | 4.53 | 22.55 |
| No | Hyperactive Type | 478 | 26.92 | 22.44 | 4.48 |
| No | Combined Type | 1268 | 44.96 | 22.49 | 22.47 |
| Unknown | No ADHD | 96 | 9.11 | 4.72 | 4.40 |
| Unknown | Inattentive Type | 82 | 26.80 | 4.39 | 22.41 |
| Unknown | Hyperactive Type | 71 | 27.15 | 22.69 | 4.46 |
| Unknown | Combined Type | 174 | 44.98 | 22.34 | 22.64 |
| Yes | No ADHD | 192 | 9.11 | 4.57 | 4.55 |
| Yes | Inattentive Type | 137 | 26.64 | 4.37 | 22.27 |
| Yes | Hyperactive Type | 115 | 27.11 | 22.39 | 4.72 |
| Yes | Combined Type | 350 | 45.04 | 22.64 | 22.40 |

| avg_anxiety | pct_within_group |
|---|---|
| Filter... | Filter... |
| 0.68 | 24.7 |
| 1.16 | 15.2 |
| 1.32 | 16.5 |
| 1.22 | 43.7 |
| 0.64 | 22.7 |
| 1.22 | 19.4 |
| 1.11 | 16.8 |
| 1.22 | 41.1 |
| 0.67 | 24.2 |
| 1.15 | 17.3 |
| 1.12 | 14.5 |
| 1.19 | 44.1 |

# 10. Severity Ranking and Outlier Detection

Patients were ranked based on their total ADHD severity scores.

Two techniques were used:

- Ranking within diagnosis groups
- Z-score based outlier detection

Patients whose severity scores were more than two standard deviations away from the mean were flagged as statistical outliers.

Output Screenshot:
*Z-score based outlier flagging using STDDEV window function*

*(only 5 tuples )*

| id | age | gender | diagnosis_label | total_severity | hyperactivity_in... |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| 2977 | 7 | Female | Combined Type | 51 | 25 |
| 2468 | 11 | Female | Combined Type | 51 | 27 |
| 4876 | 5 | Female | Combined Type | 51 | 26 |
| 3751 | 7 | Male | Combined Type | 51 | 24 |
| 3089 | 10 | Female | Combined Type | 50 | 25 |

| inattention_index | sleep_hours | daily_phone_usa... | anxiety_depressi... | severity_rank | z_score | outlier_flag |
|---|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| 26 | 6 | 4 | 0 | 1 | 2.83 | Outlier |
| 24 | 9 | 4 | 2 | 1 | 2.83 | Outlier |
| 25 | 4 | 8 | 3 | 1 | 2.83 | Outlier |
| 27 | 8 | 7 | 2 | 1 | 2.83 | Outlier |
| 25 | 6 | 4 | 3 | 5 | 2.36 | Outlier |

# 11. Comorbidity Risk Matrix

This section analyzes the co-occurrence of multiple risk factors such as:

- sleep deprivation
- high screen time
- anxiety
- learning difficulties
- organizational difficulties

The risk matrix identifies patterns where multiple risk factors appear together among ADHD patients.

Output Screenshot:
Cross-tabulation of risk factors — how they co-occur across diagnosis groups

| diagnosis_label | pct_sleep_risk | pct_high_screen | pct_anxiety_risk | pct_learning_diff | pct_org_difficulty |
|---|---|---|---|---|---|
| abc Filter... | abc Filter... | abc Filter... | abc Filter... | abc Filter... | abc Filter... |
| No ADHD | 29.2 | 31.0 | 8.7 | 22.6 | 29.8 |
| Inattentive Type | 30.2 | 28.7 | 14.0 | 25.0 | 31.9 |
| Hyperactive Type | 30.6 | 31.3 | 16.7 | 77.4 | 100.0 |
| Combined Type | 30.9 | 30.1 | 16.5 | 77.7 | 100.0 |

| pct_sleep_and_s... | pct_sleep_and_a... | pct_all_three_risks |
|---|---|---|
| abc Filter... | abc Filter... | abc Filter... |
| 9.3 | 2.1 | 0.5 |
| 7.3 | 4.6 | 2.0 |
| 10.7 | 5.4 | 1.7 |
| 9.0 | 5.5 | 1.5 |

# 12. Gender Analysis

Gender differences in ADHD prevalence were analyzed.

Two analyses were conducted:

**Gender Prevalence**

Examines ADHD distribution among male and female patients.

**Gender Severity Gap**

Calculates the difference in severity scores between genders.

These insights help understand gender-specific patterns in ADHD diagnosis.

Output Screenshot:

12a. Gender prevalence

| gender | diagnosis_label | patient_count | pct_within_gender | avg_severity | avg_hyperactivity |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| Female | No ADHD | 431 | 26.8 | 8.95 | 4.50 |
| Female | Inattentive Type | 248 | 15.4 | 27.22 | 4.63 |
| Female | Hyperactive Type | 250 | 15.5 | 27.02 | 22.42 |
| Female | Combined Type | 681 | 42.3 | 44.96 | 22.51 |
| Male | No ADHD | 575 | 22.9 | 8.98 | 4.49 |
| Male | Inattentive Type | 411 | 16.4 | 26.79 | 4.39 |
| Male | Hyperactive Type | 414 | 16.5 | 26.95 | 22.48 |
| Male | Combined Type | 1111 | 44.2 | 44.99 | 22.51 |

12b. Gender severity gap (pivot-style)

| diagnosis_label | avg_severity_male | avg_severity_fe... | severity_gap_m_... |
|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... |
| No ADHD | 8.98 | 8.95 | 0.04 |
| Inattentive Type | 26.79 | 27.22 | -0.42 |
| Hyperactive Type | 26.95 | 27.02 | -0.06 |
| Combined Type | 44.99 | 44.96 | 0.04 |

# 13. Educational Level vs ADHD

This analysis evaluates ADHD prevalence across different education levels.

Metrics include:

- ADHD prevalence percentage
- average symptom severity
- focus scores
- learning difficulty indicators

Education levels are ranked based on ADHD prevalence rates.

Output Screenshot:
*How does school stage map to diagnosis rates?*

| educational_level | total | adhd_count | adhd_prevalenc... | avg_severity | avg_focus |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| Not Working | 91 | 70 | 76.9 | 30.31 | 4.34 |
| Primary | 3259 | 2469 | 75.8 | 30.47 | 4.34 |
| Secondary | 662 | 501 | 75.7 | 30.71 | 4.31 |
| Middle | 638 | 482 | 75.5 | 29.88 | 4.33 |
| Kindergarten | 1291 | 967 | 74.9 | 30.17 | 4.52 |
| Working | 279 | 207 | 74.2 | 29.71 | 4.35 |
| University | 280 | 204 | 72.9 | 29.70 | 4.37 |

| avg_learning_diff | prevalence_rank |
|---|---|
| Filter... | Filter... |
| 0.57 | 1 |
| 0.56 | 2 |
| 0.55 | 3 |
| 0.55 | 4 |
| 0.54 | 5 |
| 0.56 | 6 |
| 0.55 | 7 |

# 14. Executive Summary Analysis

The executive summary query consolidates key metrics for stakeholders.

It summarizes:

- ADHD prevalence by life stage
- subtype distribution
- average severity scores
- lifestyle indicators
- family history percentages

This provides a simplified overview of the dataset insights.

Output Screenshot:
One-query summary a non-technical stakeholder can read

| life_stage | total_patients | adhd_prevalenc... | inattentive_count | hyperactive_count | combined_count |
|---|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... | Filter... |
| Early Childhood (3-4) | 846 | 75.1 | 143 | 133 | 359 |
| Pediatric (5-14) | 4121 | 75.6 | 659 | 664 | 1792 |
| Adolescent (15-18) | 883 | 75.8 | 140 | 152 | 377 |
| Adult (19-55) | 650 | 74.0 | 108 | 101 | 272 |

| avg_severity | avg_sleep | avg_screen_time | avg_anxiety | family_history_pct |
|---|---|---|---|---|
| Filter... | Filter... | Filter... | Filter... | Filter... |
| 30.2 | 7.5 | 3.6 | 1.1 | 21.3 |
| 30.4 | 7.5 | 3.6 | 1.1 | 19.3 |
| 30.4 | 7.6 | 3.5 | 1.1 | 22.2 |
| 29.8 | 7.6 | 3.5 | 1.1 | 20.0 |

# Key Insights

Key findings from the analysis include:

1. ADHD prevalence is highest among pediatric patients aged 5–14.
2. Individuals with higher screen time show increased symptom severity.
3. Sleep deprivation strongly correlates with higher ADHD risk categories.
4. ADHD patients report higher anxiety and learning difficulty levels.
5. Family history significantly increases the likelihood of ADHD diagnosis.

# Conclusion

This project demonstrates how advanced SQL analytics techniques can be applied to clinical datasets to uncover meaningful behavioral insights.

Using PostgreSQL analytical capabilities such as window functions, statistical aggregates, and risk modeling through CTE pipelines, the analysis reveals patterns linking ADHD diagnosis with lifestyle behaviors and mental health indicators.

Such data-driven insights can support improved early detection strategies and help researchers better understand the behavioral dynamics associated with ADHD.